Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Frank Teicheira and Son

Physical address of dairy:

Number and Street	City	County	Zip Code
27527 S Airport WAY	Manteca	San Joaquin	95337

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X257-XX04-XX14-XXXX X257-XX04-XX15-XXXX X257-XX04-XX16-XXXX X257-XX04-XX17-XXXX X257-XX04-XXX1-XXXX

B. OPERATORS

Teicheira, Frank

Operator name: Teicheira, Frank

Telephone no.: (209) 823-5844 Landline (209) 403-0638 Cellular

Mailing Address Number and Street	City	State	Zip Code
27527 S Airport WAY	Manteca	CA	95337

This operator is responsible for paying permit fees.

C. OWNERS

Teicheira, Frank

Legal owner name: Teicheira, Frank

Telephone no.: (209) 823-5844 Landline (209) 403-0638 Cellular

Mailing Address Number and Street	City	State	Zip Code
27527 S Airport WAY	Manteca	CA	95337

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	0	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation District	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/01/2023	Solid commercial fertilizer *lUrea*	13.75 *ton*	Dry-weight	1.0	46.700000	0.000000	0.000000	0.000000
06/01/2023	Liquid commercial fertilizer *UN32*	20,130.00 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	66,469.23	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	66,469.23	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
100 acre	100	100	2	none	X257-XX04-XX01-XXXX X257-XX04-XX16-XXXX X257-XX04-XX17-XXXX
45 & 15 acre	58	58	2	none	X257-XX04-XX16-XXXX
60 acre	60	60	2	none	X257-XX04-XX17-XXXX
Miller	41	41	1	none	X257-XX04-XX14-XXXX X257-XX04-XX15-XXXX
Stevens	41	41	1	none	X257-XX04-XX15-XXXX
Totals for areas that were used for application					
Totals for areas that were not used for application	300	300	8		
Land application area totals	300	300	8		

B. CROPS AND HARVESTS

100 acre

Field name: 100 acre

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 100 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,500.00 *ton*	Dry-weight		60.1	8,800.00	1,900.00	11,100.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	105.34	22.74	132.87	957.60

100 acre

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 100 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	2,580.00 *ton*	Dry-weight		62.7	4,800.00	1,000.00	4,400.00		2.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.80	92.38	19.25	84.69	423.43

45 & 15 acre

Field name: 45 & 15 acre

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 58 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	870.00 *ton*	Dry-weight		66.9	9,700.00	2,500.00	13,000.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	96.32	24.83	129.09	883.77

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	1,480.00 *ton*	Dry-weight		64.8	4,700.00	900.00	4,600.00		2.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.52	84.43	16.17	82.64	359.28

60 acre

Field name: 60 acre

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	900.00 *ton*	Dry-weight		67.2	8,500.00	2,000.00	11,600.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	83.64	19.68	114.14	984.00

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,540.00 *ton*	Dry-weight		66.3	4,400.00	700.00	4,700.00		2.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.67	76.12	12.11	81.31	363.29

Miller

Field name: Miller

01/01/2012: Almond, in shell

Crop: Almond, in shell Acres planted: 41 Plant date: 01/01/2012

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	32.80 *ton*	Dry-weight		10.0	51,400.00	5,030.00	7,330.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	0.80	104.00	16.00	80.00	0.00
Total actual harvest content	0.80	74.02	7.24	10.56	0.00

Stevens

Field name: Stevens

01/01/2012: Almond, in shell

Crop: Almond, in shell Acres planted: 41 Plant date: 01/01/2012

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	33.00 *ton*	Dry-weight		20.0	51,400.00	5,030.00	7,330.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	0.80	104.00	16.00	80.00	0.00
Total actual harvest content	0.80	66.19	6.48	9.44	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

100 acre - 10/26/2022: Oats, silage-soft dough

Field name: 100 acre

Crop: Oats, silage-soft dough — Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Urea	Solid commercial fertilizer	60.00	0.00	0.00	0.00	
Application event totals		60.00	0.00	0.00	0.00	

100 acre - 06/08/2023: Corn, silage

Field name: 100 acre

Crop: Corn, silage — Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	5.47	0.00	0.00	0.00	11,700,000.00 *gal*
Application event totals		5.47	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	60.00	0.00	0.00	0.00	
Application event totals		60.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.37	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		4.37	0.00	0.00	0.00	

100 acre - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.37	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		4.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.37	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		4.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.37	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		4.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.33	0.00	0.00	0.00	9,260,000.00 *gal*
Application event totals		4.33	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.37	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		4.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.37	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		4.37	0.00	0.00	0.00	

100 acre - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.37	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		4.37	0.00	0.00	0.00	

45 & 15 acre - 10/26/2022: Oats, silage-soft dough

Field name: 45 & 15 acre

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Urea	Solid commercial fertilizer	60.00	0.00	0.00	0.00	
Application event totals		60.00	0.00	0.00	0.00	

45 & 15 acre - 06/08/2023: Corn, silage

Field name: 45 & 15 acre

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	5.66	0.00	0.00	0.00	7,020,000.00 *gal*
Application event totals		5.66	0.00	0.00	0.00	

45 & 15 acre - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	60.00	0.00	0.00	0.00	
Application event totals		60.00	0.00	0.00	0.00	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.40	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.40	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.40	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.40	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.40	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

45 & 15 acre - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.40	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.40	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.40	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

60 acre - 10/26/2022: Oats, silage-soft dough

Field name: 60 acre

Crop: Oats, silage-soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Urea	Solid commercial fertilizer	60.00	0.00	0.00	0.00	
Application event totals		60.00	0.00	0.00	0.00	

60 acre - 06/08/2023: Corn, silage

Field name: 60 acre

Crop: Corn, silage Plant date: 06/08/2023

60 acre - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	5.47	0.00	0.00	0.00	7,020,000.00 *gal*
Application event totals		5.47	0.00	0.00	0.00	

07/07/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	60.00	0.00	0.00	0.00	
Application event totals		60.00	0.00	0.00	0.00	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.25	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.25	0.00	0.00	0.00	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.25	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.25	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.25	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.25	0.00	0.00	0.00	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.25	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.25	0.00	0.00	0.00	

60 acre - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.25	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.25	0.00	0.00	0.00	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.25	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.25	0.00	0.00	0.00	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.25	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.25	0.00	0.00	0.00	

09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	4.25	0.00	0.00	0.00	5,460,000.00 *gal*
Application event totals		4.25	0.00	0.00	0.00	

Miller - 01/01/2012: Almond, in shell

Field name: Miller

Crop: Almond, in shell

Plant date: 01/01/2012

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Miller - 01/01/2012: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	40.00	0.00	0.00	0.00	
Application event totals		40.00	0.00	0.00	0.00	

05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Miller - 01/01/2012: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Stevens - 01/01/2012: Almond, in shell

Field name: Stevens

Crop: Almond, in shell

Plant date: 01/01/2012

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

04/12/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	40.00	0.00	0.00	0.00	
Application event totals		40.00	0.00	0.00	0.00	

Stevens - 01/01/2012: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Stevens - 01/01/2012: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

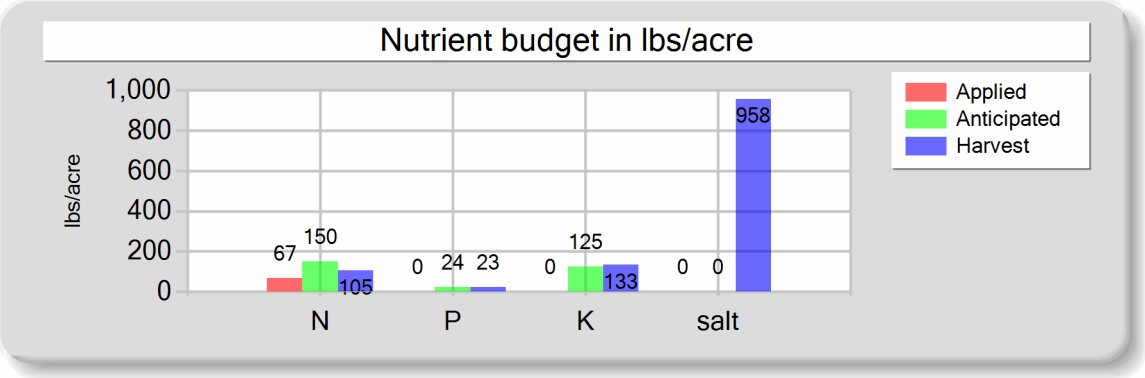
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.56	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

B. NUTRIENT BUDGET

100 acre - 10/26/2022: Oats, silage-soft dough

Field name: 100 acre | Crop: Oats, silage-soft dough | Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	60.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	67.00	0.00	0.00	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	105.34	22.74	132.87	957.60
Nutrient balance	-38.34	-22.74	-132.87	-957.60
Applied to removed ratio	0.64	0.00	0.00	0.00

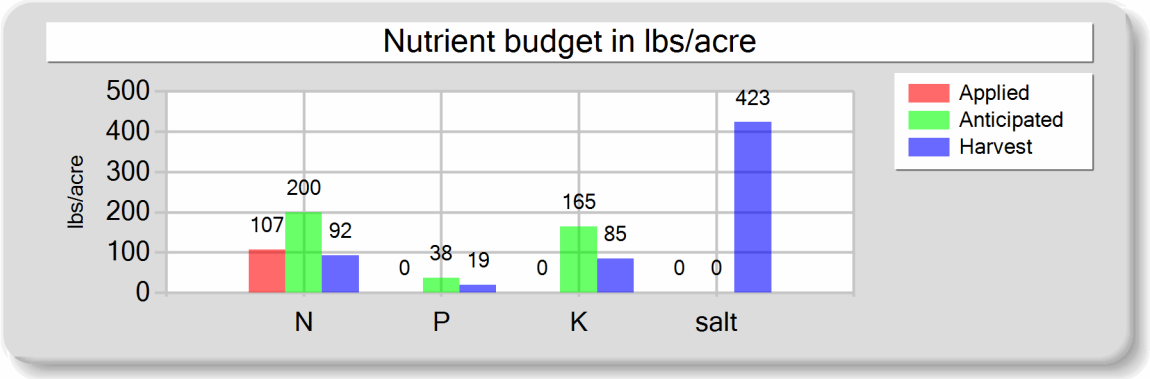
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

100 acre - 06/08/2023: Corn, silage

Field name: 100 acre Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	60.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	40.41	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	107.41	0.00	0.00	0.00
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	92.38	19.25	84.69	423.43
Nutrient balance	15.03	-19.25	-84.69	-423.43
Applied to removed ratio	1.16	0.00	0.00	0.00

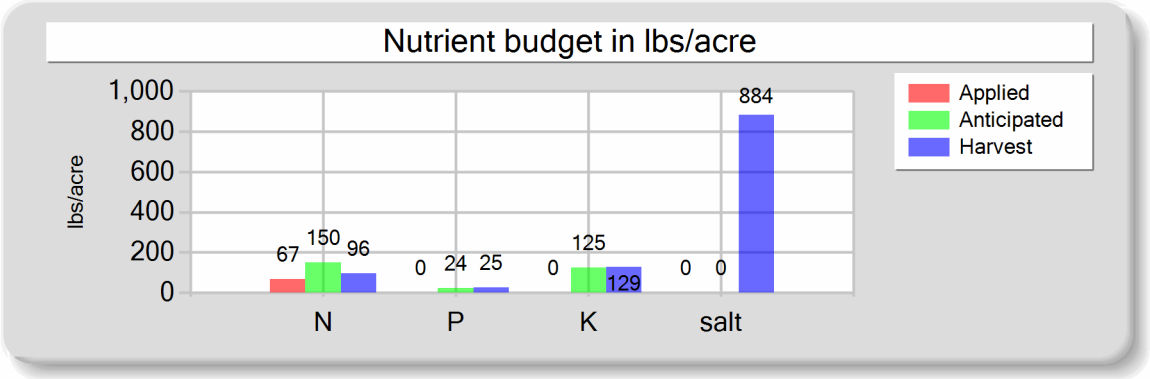
Fresh water applied	
86,480,000.00	*gallons*
3,184.76	*acre-inches*
31.85	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

45 & 15 acre - 10/26/2022: Oats, silage-soft dough

Field name: 45 & 15 acre Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	60.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	67.00	0.00	0.00	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	96.32	24.83	129.09	883.77
Nutrient balance	-29.32	-24.83	-129.09	-883.77
Applied to removed ratio	0.70	0.00	0.00	0.00

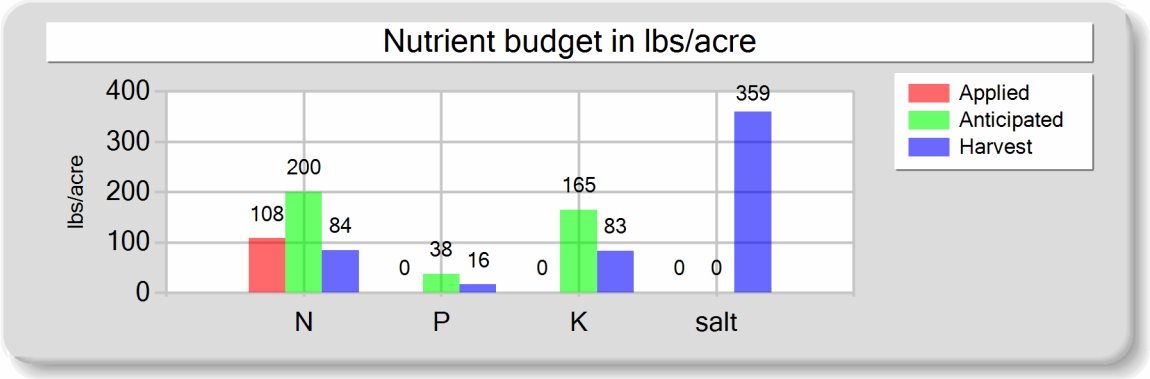
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

45 & 15 acre - 06/08/2023: Corn, silage

Field name: 45 & 15 acre Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	60.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	40.85	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	107.85	0.00	0.00	0.00
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	84.43	16.17	82.64	359.28
Nutrient balance	23.42	-16.17	-82.64	-359.28
Applied to removed ratio	1.28	0.00	0.00	0.00

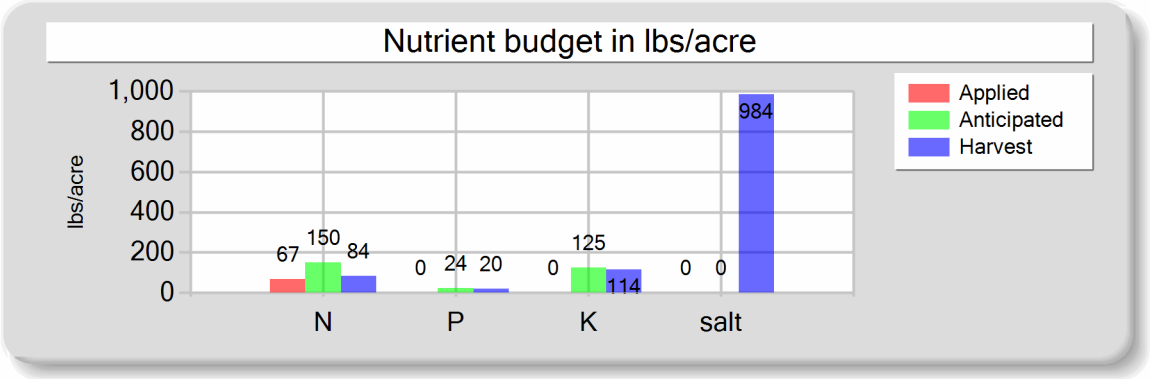
Fresh water applied	
50,700,000.00	*gallons*
1,867.11	*acre-inches*
32.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

60 acre - 10/26/2022: Oats, silage-soft dough

Field name: 60 acre | Crop: Oats, silage-soft dough | Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	60.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	67.00	0.00	0.00	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	83.64	19.68	114.14	984.00
Nutrient balance	-16.64	-19.68	-114.14	-984.00
Applied to removed ratio	0.80	0.00	0.00	0.00

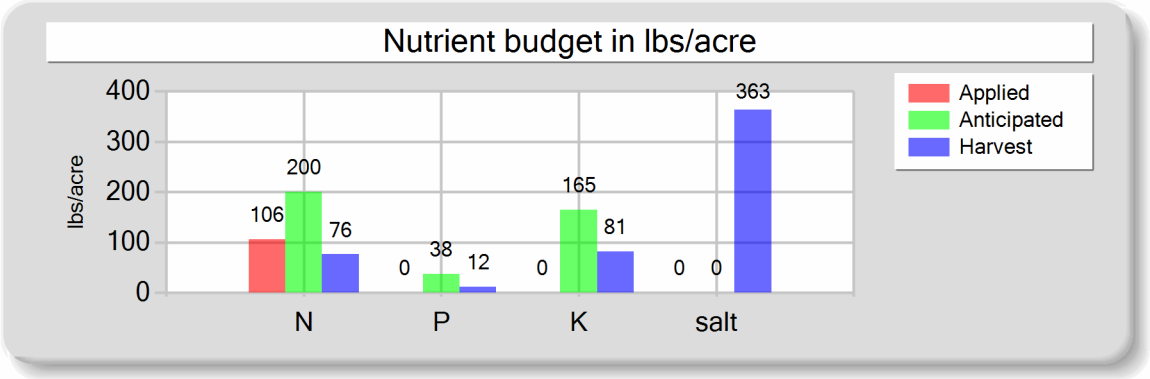
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

60 acre - 06/08/2023: Corn, silage

Field name: 60 acre Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	60.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	39.49	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	106.49	0.00	0.00	0.00
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	76.12	12.11	81.31	363.29
Nutrient balance	30.37	-12.11	-81.31	-363.29
Applied to removed ratio	1.40	0.00	0.00	0.00

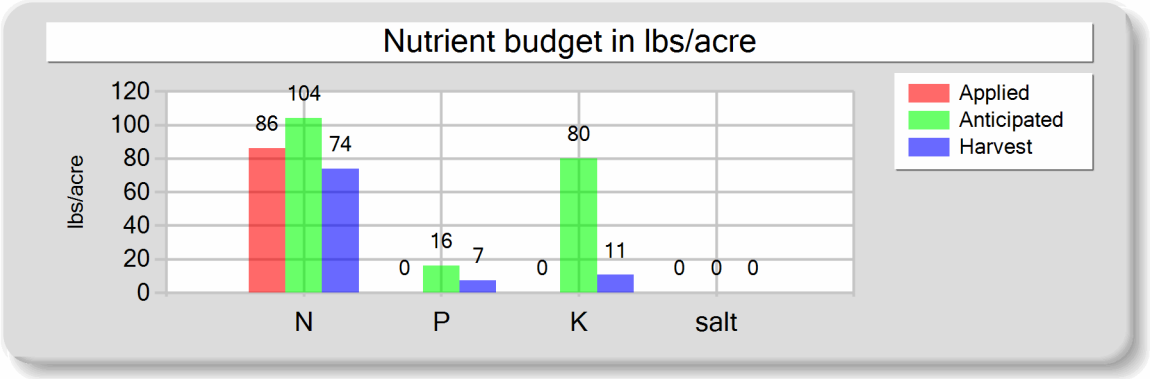
Fresh water applied	
50,700,000.00	*gallons*
1,867.11	*acre-inches*
31.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Miller - 01/01/2012: Almond, in shell

Field name: Miller | Crop: Almond, in shell | Plant date: 01/01/2012



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	32.01	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	86.01	0.00	0.00	0.00
Anticipated crop nutrient removal	104.00	16.00	80.00	0.00
Actual crop nutrient removal	74.02	7.24	10.56	0.00
Nutrient balance	11.99	-7.24	-10.56	0.00
Applied to removed ratio	1.16	0.00	0.00	0.00

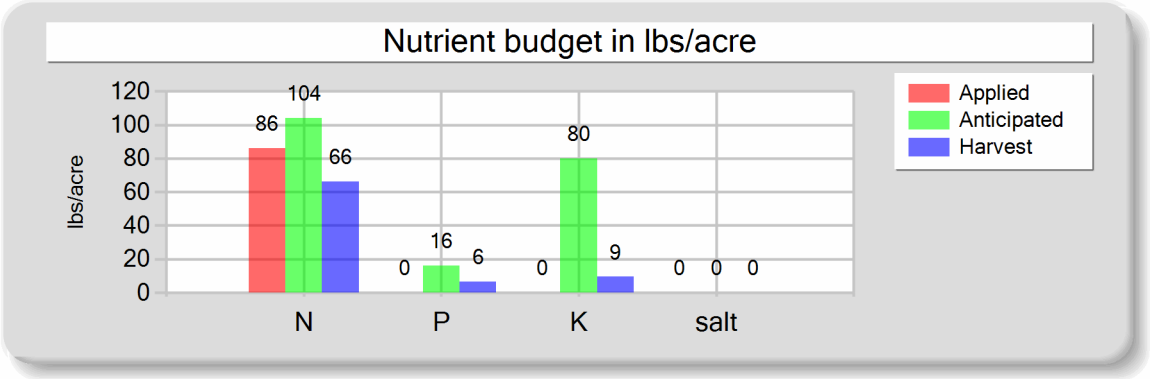
Fresh water applied	
28,080,000.00	*gallons*
1,034.09	*acre-inches*
25.22	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stevens - 01/01/2012: Almond, in shell

Field name: Stevens Crop: Almond, in shell Plant date: 01/01/2012



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	32.01	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	86.01	0.00	0.00	0.00
Anticipated crop nutrient removal	104.00	16.00	80.00	0.00
Actual crop nutrient removal	66.19	6.48	9.44	0.00
Nutrient balance	19.81	-6.48	-9.44	0.00
Applied to removed ratio	1.30	0.00	0.00	0.00

Fresh water applied	
28,080,000.00	*gallons*
1,034.09	*acre-inches*
25.22	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 01/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	13,200.00	4,640.00							
DL	100.00	100.00	10.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 08/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 3.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	7,150.00	11,600.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 01/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	116.00	61.60	0.00	0.00	65.80	724.00								3,630.00	2,840
DL	1.00	1.00	1.00	1.00	1.00	1.00								10.00	10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 08/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	33.60	11.20	0.00	0.00	35.50	297.00								179.00	1,080
DL	1.00	1.00	1.00	1.00	1.00	1.00								1.00	10

C. FRESH WATER ANALYSES

Irrigation District

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 05/04/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.60	0.00	0.00								0.00	0
DL	0.10	0.10	0.10								1.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

100 acre - 10/26/2022: Oats, silage-soft dough

100 acre - 10/26/2022: Oats, silage-soft dough

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,800.00	1,900.00	11,100.00		8.00
DL	100.00	100.00	100.00		0.10

100 acre - 06/08/2023: Corn, silage

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,000.00	4,400.00		2.20
DL	100.00	10.00	100.00		0.10

45 & 15 acre - 10/26/2022: Oats, silage-soft dough

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,700.00	2,500.00	13,000.00		8.90
DL	100.00	100.00	100.00		0.10

45 & 15 acre - 06/08/2023: Corn, silage

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 10/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	900.00	4,600.00		2.00
DL	100.00	10.00	100.00		0.10

60 acre - 10/26/2022: Oats, silage-soft dough

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 05/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,500.00	2,000.00	11,600.00		10.00
DL	100.00	100.00	100.00		0.10

60 acre - 06/08/2023: Corn, silage

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	700.00	4,700.00		2.10
DL	100.00	10.00	100.00		0.10

Miller - 01/01/2012: Almond, in shell

Estimate - no laboratory analysis

Sample and source description: Estimate - no laboratory analysis

Sample date: 10/20/2023 Source of analysis: Other/estimated Method of reporting: Dry-weight

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	5,030.00	7,330.00		
DL	10.00	10.00	10.00		

Stevens - 01/01/2012: Almond, in shell

Estimate - no laboratory analysis

Sample and source description: Estimate - no laboratory analysis

Sample date: 10/20/2023 Source of analysis: Other/estimated Method of reporting: Dry-weight

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	5,030.00	7,330.00		
DL	10.00	10.00	10.00		

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

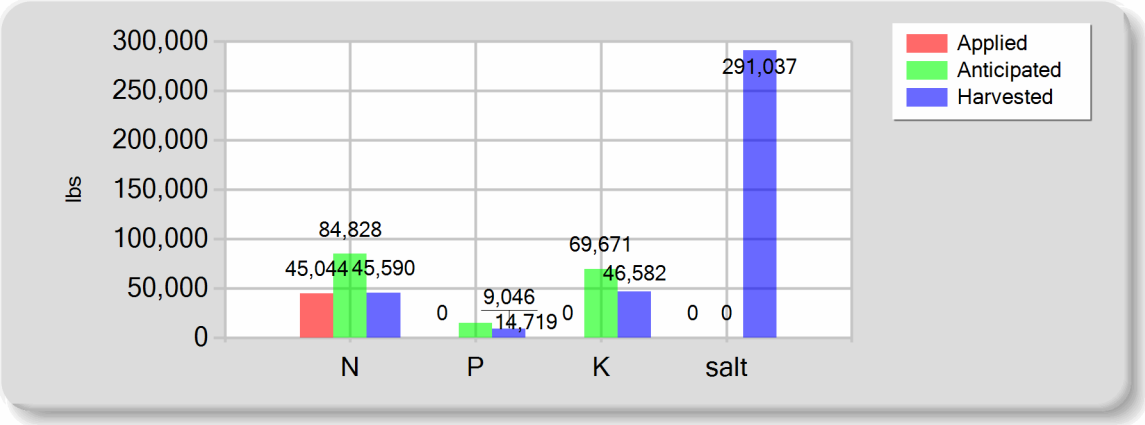
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

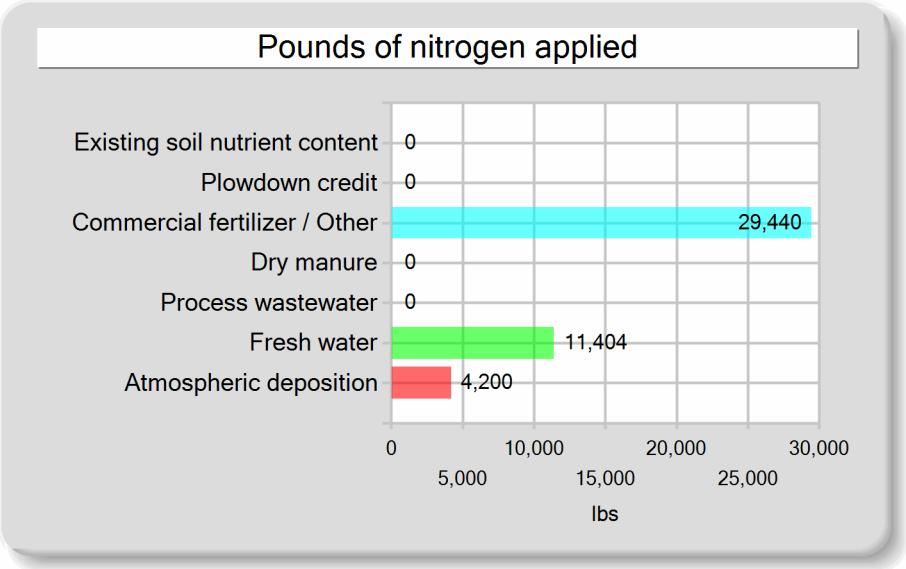
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

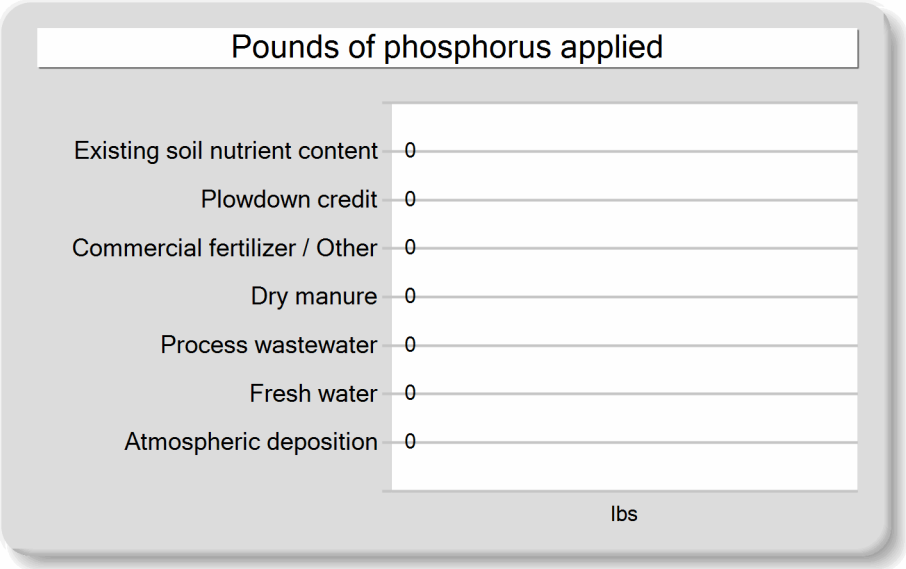
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	29,440.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11,404.48	0.00	0.00	0.00
Atmospheric deposition	4,200.00	0.00	0.00	0.00
Total nutrients applied	45,044.48	0.00	0.00	0.00
Anticipated crop nutrient removal	84,828.00	14,719.00	69,671.00	0.00
Actual crop nutrient removal	45,589.71	9,046.49	46,582.18	291,037.18
Nutrient balance	-545.23	-9,046.49	-46,582.18	-291,037.18
Applied to removed ratio	0.99	0.00	0.00	0.00

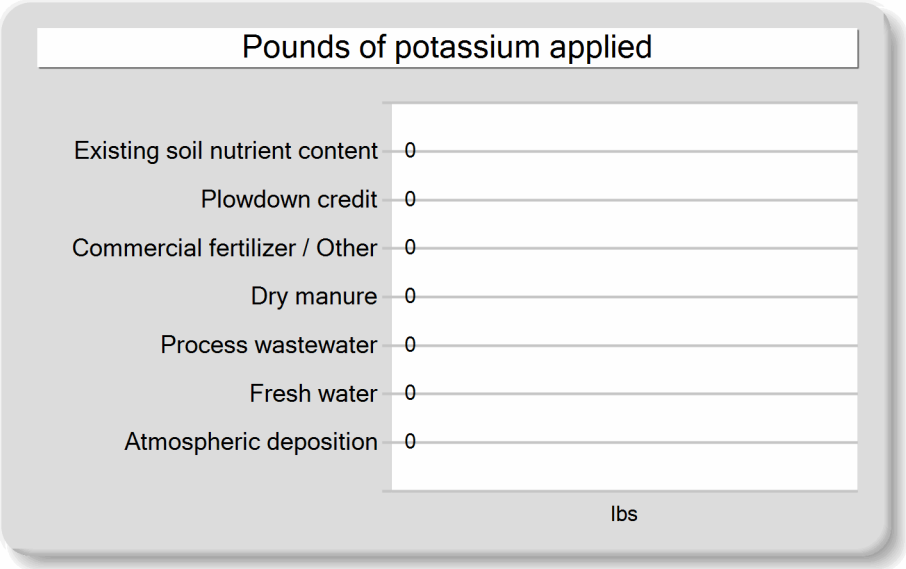
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

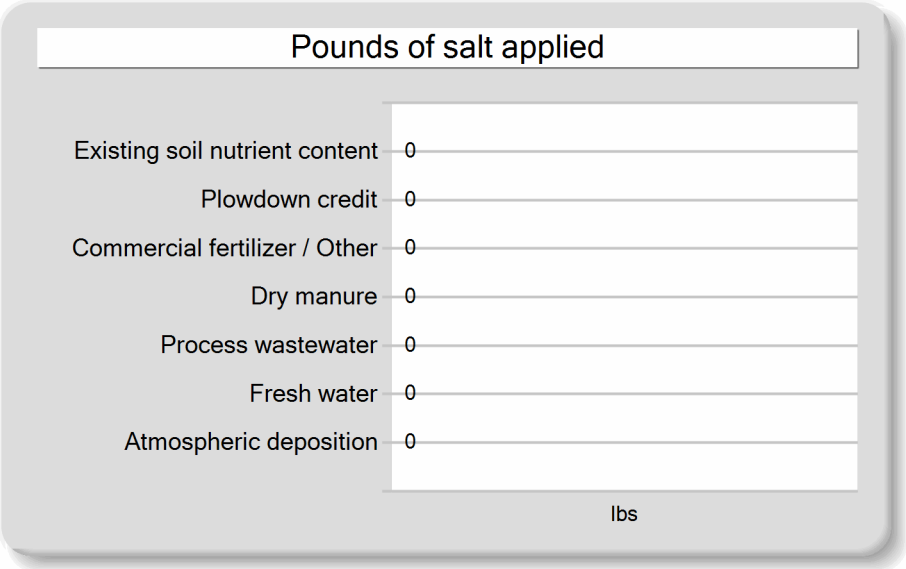


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Almonds were not analyzed by a lab. N, P, & K estimated.
No cattle on the dairy.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Teicheira	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Frank Teicheira

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Teicheira	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-18-2024	
DATE	DATE

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Dutra Farms
5496 W. Ripon Rd.
Manteca , CA 95336

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER:
RECEIVED DATE: 5/4/23 3:17 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 5/8/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 18.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irrigation District Water	T3124033A	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	< 10.0	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	5.60	mg/L
		Soluble Salts (SALT-SOL)	Calculation	55.7	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101			4.19 (T3123068J)	4.18 (Dup)	0.239
Electrical Conductivity (EC)	100					5.55
Nitrate Nitrogen (NO3-N)	102	100 (T3123010A)	106 (T3123010A)			
Total Dissolved Solids (TDS)	103			0.000 (T3124034A)	0.000 (Dup)	
Total Nitrogen (TN)	96.3					

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DUTRA FARMS INC
Mailing address: 5496 WEST RIPON ROAD
(not facility address) MANTECA CA 95337
Phone#: 209-479-4502

Authorized Copy Released To: BRUCE GOLDSMITH

Sampled By: BRUCE GOLDSMITH
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	5-4-23	1:50	IRRIGATION DISTRICT	WATER	18.0°C						H7
2											
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[illegible]	DAIRY COMPLIANCE GROUP				
2nd		DENELE ANALYTICAL, INC.	5-4-23 3:17			T3124033

Required ~~to be~~ submitted to any agency? (YES / NO)

IEH - JL ANALYTICAL

217 Primo Way • Modesto, California 95358 • Office (209) 538-8111 • FAX (209) 538-3966

Frank Teicheira & Son

27527 South Airport Way
Manteca,CA 95337

Report # L2.2-14R66607
Report Date: 02/05/2024
Received Date: 01/25/2024
Work Order: 626264

Laboratory Number: 62626401
Description: Water, Frank Well, 01/25/24 0950
Sampled By:Frank Teicheira, Employed By:Frank Teicheira + Son

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Ammoniacal Nitrogen by Kjeldahl	< 2	mg/l	2	0.3	SM 4500 NH3 C	01/25/2024	11:50	JN
Electrical Conductivity (E.C.)	470	umho/cm	1	1	SM 2510B[5]	02/02/2024	13:20	KB
Nitrate Nitrogen	< 0.4	mg/l	0.4	0.01	EPA 300.0	01/25/2024	12:32	KB

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Response %
Ammoniacal Nitrogen by Kjeldahl	mg/l	< 2	< 2	28	27	27	27	0.5	97.4	96.6
Electrical Conductivity (E.C.)	umho/cm	< 1	470	1400	1800	1800	1400	0.2	93.8	100.1
Nitrate Nitrogen	mg/l	< 0.1	10	0.9	11	11	0.9	0.1	96.9	96.8

Notes

5 At 25 °C

Authorized By: _____ 02/05/2024

Britney Dundee, Chemistry Supervisor

ELAP Accreditation Laboratory Certificate #2776. Unless otherwise noted, all samples were received in acceptable condition. The result(s) in this report relate only to the portion of the sample(s) tested. Results are not corrected for recovery or expanded measurment of uncertainty. This report does not constitute a release of product for consumption. This report shall not be reproduced except in full, without written approval of the laboratory. This document contains confidential commercial information pursuant to 5 U.S.C. SEC. 552(b)(4).

IEH - JL ANALYTICAL

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Frank Teicheira & Son

27527 South Airport Way
Manteca,CA 95337

Report # L2.2-14R66607
Report Date: 02/05/2024
Received Date: 01/25/2024
Work Order: 626264

Laboratory Number: 62626402
Description: Water, Barn Well Sow, 01/25/24 0940
Sampled By:Frank Teicheira, Employed By:Frank Teicheira + Son

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Ammoniacal Nitrogen by Kjeldahl	< 2	mg/l	2	0.3	SM 4500 NH3 C	01/25/2024	11:50	JN
Electrical Conductivity (E.C.)	410	umho/cm	1	1	SM 2510B[5]	02/02/2024	13:20	KB
Nitrate Nitrogen	< 0.4	mg/l	0.4	0.01	EPA 300.0	01/25/2024	12:32	KB

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Response %
Ammoniacal Nitrogen by Kjeldahl	mg/l	< 2	< 2	28	27	27	27	0.5	97.4	96.6
Electrical Conductivity (E.C.)	umho/cm	< 1	470	1400	1800	1800	1400	0.2	93.8	100.1
Nitrate Nitrogen	mg/l	< 0.1	10	0.9	11	11	0.9	0.1	96.9	96.8

Notes

5 At 25 °C

Authorized By: _____ 02/05/2024

Britney Dundee, Chemistry Supervisor

ELAP Accreditation Laboratory Certificate #2776. Unless otherwise noted, all samples were received in acceptable condition. The result(s) in this report relate only to the portion of the sample(s) tested. Results are not corrected for recovery or expanded measurment of uncertainty. This report does not constitute a release of product for consumption. This report shall not be reproduced except in full, without written approval of the laboratory. This document contains confidential commercial information pursuant to 5 U.S.C. SEC. 552(b)(4).

IEH - JL ANALYTICAL

217 Primo Way • Modesto, California 95358 • Office (209) 538-8111 • FAX (209) 538-3966

Frank Teicheira & Son

27527 South Airport Way
Manteca,CA 95337

Report # L2.2-14R66607
Report Date: 02/05/2024
Received Date: 01/25/2024
Work Order: 626264

Laboratory Number: 62626403
Description: Water, Frank Domestic Well, 01/25/24 0945
Sampled By:Frank Teicheira, Employed By:Frank Teicheira + Son

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Ammoniacal Nitrogen by Kjeldahl	< 2	mg/l	2	0.3	SM 4500 NH3 C	01/25/2024	11:50	JN
Electrical Conductivity (E.C.)	430	umho/cm	1	1	SM 2510B[5]	02/02/2024	13:20	KB
Nitrate Nitrogen	< 0.4	mg/l	0.4	0.01	EPA 300.0	01/25/2024	12:32	KB

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Response %
Ammoniacal Nitrogen by Kjeldahl	mg/l	< 2	< 2	28	27	27	27	0.5	97.4	96.6
Electrical Conductivity (E.C.)	umho/cm	< 1	470	1400	1800	1800	1400	0.2	93.8	100.1
Nitrate Nitrogen	mg/l	< 0.1	10	0.9	11	11	0.9	0.1	96.9	96.8

Notes

5 At 25 °C

Authorized By: _____ 02/05/2024

Britney Dundee, Chemistry Supervisor

ELAP Accreditation Laboratory Certificate #2776. Unless otherwise noted, all samples were received in acceptable condition. The result(s) in this report relate only to the portion of the sample(s) tested. Results are not corrected for recovery or expanded measurment of uncertainty. This report does not constitute a release of product for consumption. This report shall not be reproduced except in full, without written approval of the laboratory. This document contains confidential commercial information pursuant to 5 U.S.C. SEC. 552(b)(4).

Company Name: FRANK Teicheira + Sons
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Phone: (209) 403-4638
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JL Analytical Services
217 Primo Way
Modesto, CA 95358
Phone (209) 538-8111
FAX (209) 538-3966

2F077 ver. 3 ELAP chain of custody

Analysis Requested (write in the tests)

Lab Use Only: Work Order Number	Matrix	Sample Information: Sample Description Annual	Sampling Date	Sampling Time	Chlorine Level	Field pH	Sample Type	Reason For Analysis	Sample Preserv.	E. C.	Ammonium Nitrogen	Nitrate-Nitrogen						Thermometer ID: ____	Temp as read	Corr. Temp
										Check the Appropriate Box Below:										
	DW	Frank House	3/2/23	[illegible]			W	A	NONE	X	X	X								
		Barn		1335			W	A		X	X	X								
		Virginia		1340			W	A		X	X	X								

Sampled by: Frank Teicheira				
Relinquished by (signature): Frank Teicheira	Received by (signature):	Date: 3/2/23	Time: 1457	
Relinquished by (signature):	Received by (signature):	Date:	Time:	

Key:

Sub sample bottle ID: If more than one bottle per sample assign a unique letter to each (A, B, C, etc.)

Matrix: DW=Drinking Water GW=Ground Water SW=Storm Water WW=Waste Water

Sample Type: D=Distribution W=Well O=Other

Reason for Analysis: A=Routine B=Repeat C=Special D=Other

Sample Preserv.: M: Cool, <10 deg C, Sodium Thiosulfate (Na2S2O3), C: </= 6 deg C NA: Nitric Acid (HNO3) SA: Sulfuric Acid (H2SO4) H: Hydrochloric Acid (HCl)

Notes / Special Instructions:
Dairy Quality